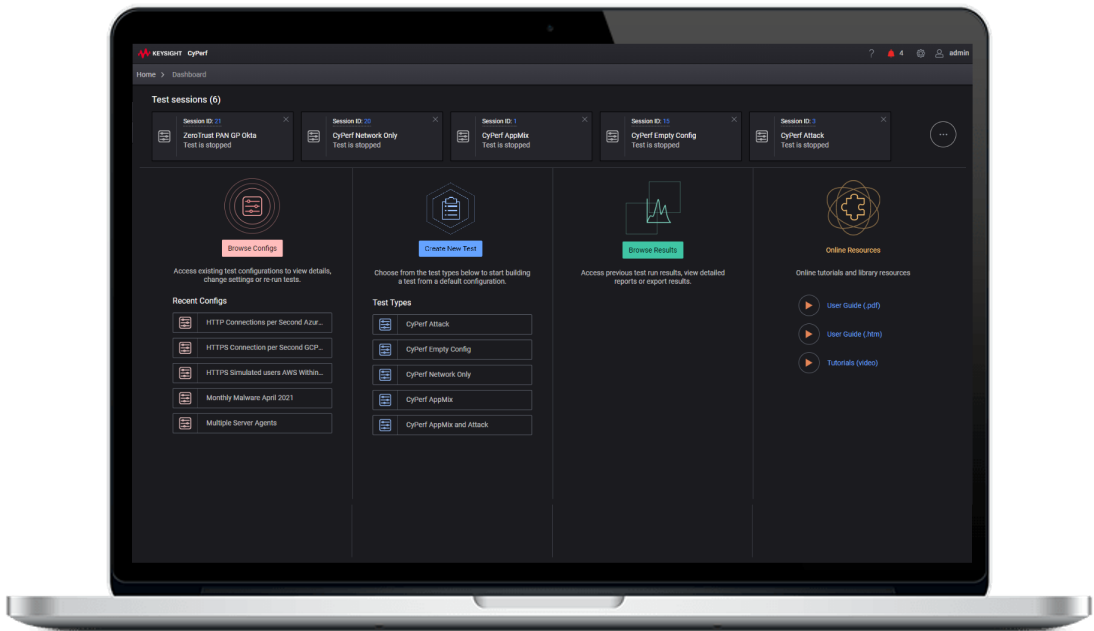


Keysight CyPerf

Release 26.0.0

Deployment Guide



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Contact us

Keysight headquarters

1400 Fountaingrove Parkway

Santa Rosa, CA 95403-1738

Email address: support@keysight.com

Website: <https://support.keysight.com/>

Support

Location	Phone number	Local time
<i>Americas</i>		
US, Canada	1-888-829-5558	8h00 – 17h00
Brazil	0800-892-0522	8h00 – 17h00
Mexico	001-888-829-5558	8h00 – 17h00
Other	+1-719-273-6516	8h00 – 17h00
<i>EMEA</i>		
Belgium	0800-18686	8h30 – 17h30
Finland	0800-913-352	8h30 – 17h30
France	0800-917228	8h30 – 17h30
Germany	0800-0824099	8h30 – 17h30
India	1800-18-02552	8h30 – 17h30
Ireland	1800-949245	8h30 – 17h30
Israel	1-809-454975	8h30 – 17h30
Italy	0800-790571	8h30 – 17h30
Luxembourg	0800-25112	8h30 – 17h30
Netherlands	0800-022-9086	8h30 – 17h30

Romania	0213 015 699	8h30 – 17h30
Spain	800-654386	8h30 – 17h30
Sweden	0201-202266	8h30 – 17h30
United Kingdom	0800-0293882	8h30 – 17h30
<i>Asia and Australia</i>		
Australia	1-800-370-558	8h30 – 17h00
China Mainland	800-810-0005	8h30 – 17h30
	400-810-0005	8h30 – 17h30
Hong Kong	800-931-613	9h00 – 18h00
Japan	0120-421-621	9h00 – 17h30
Malaysia	1800-819 092	8h30 – 17h30
South Korea	080-770-0800	8h30 – 17h30
Singapore	800-101-3797	8h30 – 17h30
Taiwan	0800-699-880	9h00 – 18h00
Other	+65 6215 7600	8h30 – 17h30 (Singapore)

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Secure Software Design Framework (SSDF)

This release is in compliance with Secure Software Development Framework (SSDF).

This release is not yet in compliance with Secure Software Development Framework (SSDF).

Keysight complies with multinational regulations for the cybersecurity of its own products and is committed to providing information to assist you in protecting your products and solutions from external cyber threats.

This section guides you on how to:

- [Keep your system secure](#)
- [Report vulnerabilities and receive fixes](#)
- [Review Keysight security management](#)
- [Securely decommission your system](#)

To sign up for Support portal access, go to <https://support.keysight.com/> and select **Register**.

Security advisories

To review security advisories and keep your system secure:

1. Access the Support portal at <https://support.keysight.com/>.
2. Select **Network Tests and Visibility Support (NAS)**.
3. Select the tile labeled **Product & Solution Cyber Security**.
The Product & Solution Cyber Security web page appears.
4. Select the tile labeled **Security Advisories**.
The Security Advisories web page appears.
5. Review and follow the instructions in the relevant security advisories.
6. If desired, download advisories.

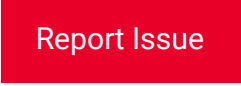
Report a vulnerability or security issue

To report a vulnerability or security issue:

1. Access the Support portal at <https://support.keysight.com/>.
2. Select **Network Tests and Visibility Support (NAS)**.
3. Select the tile labeled **Product & Solution Cyber Security**.
The Product & Solution Cyber Security web page appears.
4. Below the images, select the link at the end of the sentence in red text that says:
"To report a cyber security issue on a Keysight product please go [here](#)".

The Product and Solution Cyber Security web page appears.

5. Select either the red text hyperlink labeled "**Report an Issue**", or scroll down and select the red **Report Issue** button:



Report Issue

After you report the issue, Support works with you to resolve it. If you need a software fix, Support provides it.

Keysight security management

To review Keysight security management:

1. Access the Support portal at <https://support.keysight.com/>.
2. Select **Network Tests and Visibility Support (NAS)**.
3. Select the tile labeled **Product & Solution Cyber Security**.
The Product & Solution Cyber Security web page appears.
4. Below the images, select the link at the end of the sentence in red text that says:
"**To report a cyber security issue on a Keysight product please go [here](#)**".
The Product and Solution Cyber Security web page appears.
5. Scroll to the bottom of the web page, and under **Additional Information**, select the red text hyperlink labeled "**Keysight Security**".

This opens a PDF on Keysight global security policy for all products. It includes information on:

- Security management
- Product and solution security
- Information security
- Government security
- Supply chain security
- Physical and site security
- Data privacy policy
- Enterprise risk management
- Additional resources

Decommission a system

Because each product line is different, contact Support for product-specific instructions on how to decommission a system.

CHAPTER 1

Overview

This Deployment Guide provides step-by-step instructions for deploying Keysight CyPerf.

Keysight CyPerf consists of the following components, which can be deployed on-premises, on AWS, Azure, GCP, or Kubernetes, or in Docker container environments:

- **Keysight CyPerf Controller:** a web-based user interface for configuring and running tests, viewing real-time statistics and reviewing results. For details, see [Deploying the CyPerf Controller from OVA](#).
- **Keysight CyPerf Traffic Agent:** software agents generating test traffic. For details, see [Deploying the CyPerf Agent from OVA](#).
- **Keysight CyPerf Controller Proxy:** ensures communication between agents and controller in hybrid setups. For details, see [Deploying the CyPerf Controller Proxy from OVA](#).

Keysight CyPerf License and User Manager is a centralized license server that can be used for multiple CyPerf controller setups. For details, see [Deploying the License and User Manager from OVA](#).

Supported Deployment Platforms

A complete list of deployment platforms supported by CyPerf can be found at [CyPerf supported platforms](#).

Keysight CyPerf Agent

Keysight CyPerf agent is a primary traffic generation node used by Keysight CyPerf. It is capable of simulating stateful web traffic with or without Transport Layer Security (TLS) encryption, as is the case of banking or web portal applications. Moreover, Keysight CyPerf agent can generate simple HTTP/ HTTPS upload and download commands with real payload files. It can also generate most updated attack traffic published by OWAPS, and it allows users to configure combinations of legitimate traffic and attack traffic more realistically in order to test their DUT.

Costs and Licenses

You are responsible for the cost of the AWS, Azure, or GCP services used while running this Quick Start reference deployment. There is no additional cost for using the Quick Start.

The AWS CloudFormation templates, Azure Resource Manager templates, and GCP Deployment Manager templates for this Quick Start include configuration parameters that you can customize. Some of these settings (such as the instance type) will affect the cost of the deployment. For cost estimates, see the pricing pages for each cloud service you are using. Prices are subject to change.



NOTE: After you deploy the Quick Start, we recommend that you enable the [Cost and Usage Reports](#) available on specific clouds to track costs associated with the Quick Start. These reports provide cost estimates based on usage throughout each month and finalize the data at the end of the month. For more information about the reports, see the [specific cloud documentation](#).

CHAPTER 2

OVA Manual Deployment

This chapter details the requirements and step-by-step instructions needed to deploy each CyPerf component from an OVA image.

Deploying the CyPerf Controller from OVA

ESXi resource requirements:

- 8 CPU cores
- 16 GB of RAM
- At least 100 GB of SSD storage with thin (i.e. dynamic) provisioning, and 250 GB of SSD storage otherwise
- 1 available physical NIC on the ESXi for management connectivity

① **NOTE:** 6.5 or newer ESXi versions are supported.

To deploy the Keysight CyPerf controller from an OVA file:

1. Download the OVA from the provided location.
2. Open your preferred ESXi client and start the deployment.
3. Before starting the VM, make sure that the Management interface of the VM has environment connectivity. Note that DHCP is the preferred option, and this is also how the VM is configured to obtain its IP address. **To enable DHCP**, enter the following command: `kcov networking ip set mgmt0 dhcp`.

① **NOTE:** The application has a predefined IP address that will change during the DHCP negotiation step. This IP address will be the one configured to access the application.

To configure a static IP:

- a. Connect to the ESXi interface.
- b. Enter user `console`, and you will be redirected to a new login form.
- c. Enter the following predefined credentials: username: `admin`, password: `CyPerf&Keysight#1`

- d. To apply a static IP, enter the following command:

```
kcos networking ip set mgmt0 DESIRED_STATIC_IP/PREFIX GATEWAY_IP.
```

For example: `kcos networking ip set mgmt0 10.38.166.89/24
10.38.166.1`

4. Go to **`https://<obtainedIpAddress>`** to access the application.

Deploying the CyPerf Agent from OVA

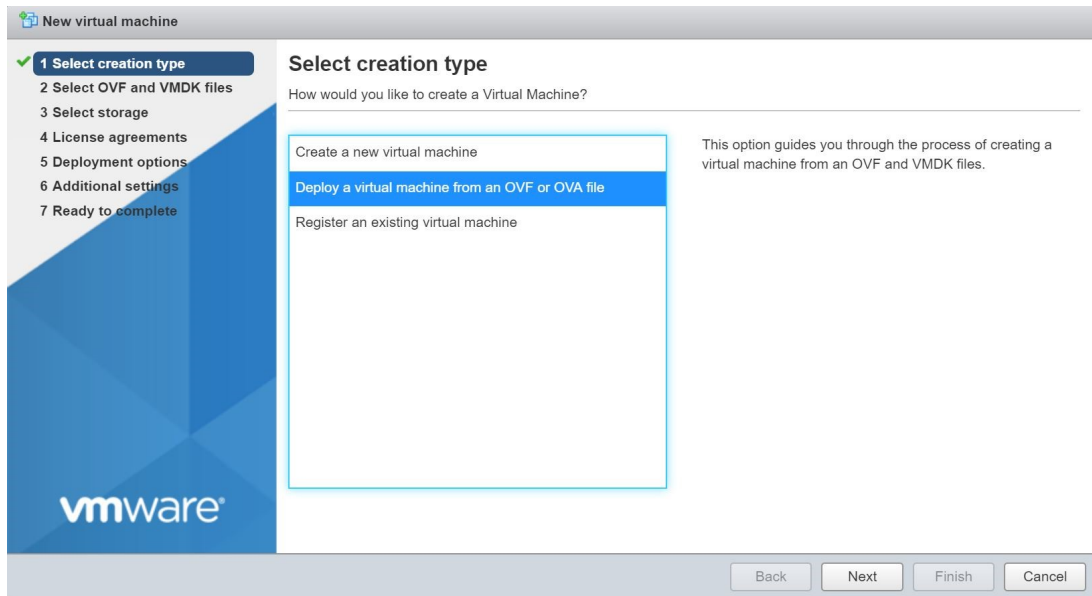
This section describes how to deploy the Keysight CyPerf agent OVA from the vSphere web interface.

**NOTE:**

- The supported vSphere version is 6.5 or newer.
- Prior to deployment, you must have a valid vSphere ESXi host with valid credentials and a VMware license.

Step 1: Deploy a VM from OVA

In the **New Virtual Machine** window, select **Create a virtual machine from an OVF or OVA file** as the creation type.



Step 2: Select OVA

Specify the **virtual machine name** and browse for the CyPerf agent OVA.

The screenshot shows the 'New virtual machine - CyPerf 1.0 Agent' wizard. On the left, a sidebar lists seven steps: 1 Select creation type, 2 Select OVF and VMDK files (highlighted), 3 Select storage, 4 License agreements, 5 Deployment options, 6 Additional settings, and 7 Ready to complete. The main area is titled 'Select OVF and VMDK files' and contains the instruction 'Select the OVF and VMDK files or OVA for the VM you would like to deploy'. Below this, there is a text input field for 'Enter a name for the virtual machine.' with the text 'CyPerf 1.0 Agent' entered. A note states: 'Virtual machine names can contain up to 80 characters and they must be unique within each ESXi instance.' A large light blue box with the text 'Click to select files or drag/drop' is centered in the main area. At the bottom right, there are four buttons: 'Back', 'Next', 'Finish', and 'Cancel'.

Step 3: Select Datastore

Indicate the storage type and datastore.

The screenshot shows the 'New virtual machine - CyPerf 1.0 Agent' wizard at Step 3: Select storage. The sidebar on the left shows steps 1 through 7, with '3 Select storage' highlighted. The main area is titled 'Select storage' and contains the instruction 'Select the datastore in which to store the configuration and disk files.' Below this, a note states: 'The following datastores are accessible from the destination resource that you selected. Select the destination datastore for the virtual machine configuration files and all of the virtual disks.' A table lists the available datastores:

Name	Capacity	Free	Type	Thin pro...	Access
datastore1	3.63 TB	2.01 TB	VMFS6	Supported	Single

At the bottom right of the table, it says '1 items'. At the bottom of the wizard, there are four buttons: 'Back', 'Next' (highlighted), 'Finish', and 'Cancel'.

Step 4: Select the Network

Agents must have two interfaces: a management interface and a test interface.

1. Specify **VM Network** as the Management Network.
2. From the drop-down list, select a proper vSwitch from the drop-down list which has LAN network connectivity.

In the following example, **VM Network** is LAN vSwitch.

3. Specify **Test_net1** as the Test Network.
4. From the drop-down list, select a proper vSwitch which has test network connectivity.
In the following example, **Test_net1** also selects LAN vSwitch. This is because the user opts for LAN IP for both Management and Test network.
5. Select vSwitches as per your test topology.
6. Select **Next** and start deployment.

New virtual machine - CyPerf 1.0 Agent

✓ 1 Select creation type
✓ 2 Select OVF and VMDK files
✓ 3 Select storage
✓ 4 Deployment options
5 Ready to complete

Deployment options

Select deployment options

Network mappings	Management Network: VM Network
	Test Network: Test_net1
Disk provisioning	☒ Thin ☐ Thick

Back Next Finish Cancel

Step 5: Set *netplan*

After the deployment is complete and the VM is in running state, the management interface must get an IP from the DHCP server.

1. Establish a SSH connection to the CyPerf agent using the following credentials:
 - username: **cyperf**
 - password: **cyperf**.
2. Configure the netplan settings in the file **/etc/netplan/01-netcfg.yaml** to set the IP address of the test interface. Choose either DHCP or static configuration based on your requirements. For more details, see the [netplan documentation](#).

NOTE: If the Management and Test Network are in the same subnet, then the management interface must have a lower metric than the test interface.

In the following example, the user selects the same VM Network for the test and management subnet. **ens33** is the test interface, and **ens160** is the management interface. That is why the user sets metric **100** for the management interface and metric **200** for the test interface in netplan.

Example:

```
network:
  version: 2
  renderer: networkd
  ethernets:
    ens33:
      dhcp4: yes
      dhcp-identifier: mac
      dhcp4-overrides:
        route-metric: 200
    ens160:
      dhcp4: yes
      dhcp-identifier: mac
      dhcp4-overrides:
        route-metric: 100
```

3. After netplan configuration, apply netplan:

```
sudo netplan apply
```

4. Set the netplan configuration for a static IP for both the management and test interfaces:

```
network:
  version: 2
  renderer: networkd
  ethernets:
```

```

ens160:
  dhcp4: no
  addresses: [IP address/Netmask]
  routes:
    - to: 0.0.0.0/0
      via: <Gateway IP>
      metric: 100
ens192:
  dhcp4: no
  addresses: [IP address/Netmask]
  routes:
    - to: 0.0.0.0/0
      via: <Gateway IP>
      metric: 200

```

5. Set the netplan configuration for a static IP in the management interface, and with no IP for the test interface:

```

network:
  version: 2
  renderer: networkd
  ethernets:
    ens160:
      dhcp4: no
      addresses: [IP address/Netmask]
      routes:
        - to: 0.0.0.0/0
          via: <Gateway IP>
          metric: 100
    ens192:
      dhcp4: no

```

Step 6: Set *cyperfagent* configuration

Use the CLI *cyperfagent* to configure the required parameters.

The following commands are available in CLI:

- `cyperfagent controller show`: Shows the currently configured controller.
- `cyperfagent controller set <Controller-IPv4|Controller-Hostname> <[--interactive]>|<[--username=<USERNAME> --password=<PASSWORD>] [--fingerprint=<FINGERPRINT>]|[--skip-identity-verification]> | [flags]`: Sets the controller.
For more details, see [Connecting CyPerf Agents to a Controller](#).
- `cyperfagent interface management show`: Shows the currently configured Management Interface.
- `cyperfagent interface management set ens160`: Sets the Management Interface to `ens160`.
- `cyperfagent interface management set auto`: Sets the Management Interface to `auto` for dynamically detecting the Interface at runtime.
- `cyperfagent interface test set ens160`: Sets the Test Interface to `ens160`.
- `cyperfagent interface test set auto`: Sets the Test Interface to `auto` for dynamically detecting the Interface at runtime.

Step 7: Disable the *Synchronize guest time with host* option

NOTE: Keysight CyPerf agents deployed by agent OVA should have the **Synchronize guest time with host** option disabled by default. You can enable this option for the EXSi Hypervisor.

1. Under the **Agent VM for vSphere** web interface, select **VM Options**.
2. Under **VMware Tools > Time**, disable **Synchronize guest time with host**.

The screenshot shows the 'VMware Tools' configuration page. The 'Time' section is highlighted with a red box around the 'Synchronize guest time with host' checkbox, which is currently unchecked. Other sections visible include 'Power Operations', 'Run VMware Tools Scripts', 'Tools Upgrades', and 'Power management'.

Power Operations	Shut Down Guest Put Guest on Standby Power On / Resume VM Restart Guest
Run VMware Tools Scripts	☒ After powering on ☒ After resuming ☒ Before suspending ☒ Before shutting down guest
Tools Upgrades	☒ Check and upgrade VMware Tools before each power on
Time	☐ Synchronize guest time with host
Power management	Standby response How should the virtual machine respond when the guest OS is placed on standby? ☒ Suspend the virtual machine.

Deploying the CyPerf Controller Proxy from OVA

ESXi resource requirements:

- 2 CPU cores
- 2 GB of RAM
- At least 10 GB of SSD storage
- 1 available physical NIC on the ESXi for management connectivity

 **NOTE:** 6.5 or newer ESXi versions are supported.

To deploy the CyPerf controller proxy from an OVA file:

1. Download the OVA from the provided location.
2. Open your preferred ESXi client and start the deployment.
3. Before starting up the VM, make sure that the management interface of the VM has environment connectivity. Note that DHCP is the preferred option, and this is also how the VM is configured to obtain its IP address.
4. CyPerf agents should also be deployed with the controller proxy. CyPerf controller connects to the obtained IP address of the CyPerf controller proxy on port 443 and accesses the CyPerf agents through it.

Deploying the License and User Manager from OVA

ESXi resource requirements:

- 2 CPU cores
- 4 GB of RAM
- At least 50 GB of SSD storage
- 1 available physical NIC on the ESXi for management connectivity

 **NOTE:** 6.5 or newer ESXi versions are supported.

To deploy the Keysight License and User Manager from an OVA file:

1. Download the OVA from the provided location.
2. Open your preferred ESXi client and start the deployment.
3. Before starting up the VM, make sure that the management interface of the VM has environment connectivity. Note that DHCP is the preferred option, and this is also how the VM is configured to obtain its IP address.
4. Go to **<https://<obtainedIpAddress>>** to access the application. Enter credentials: **admin/CyPerf&Keysight#1**

You will now be able to activate licenses and use the License and User Manager as the external license server on your CyPerf controller setup.

CHAPTER 3

QCOW2 Deployment on KVM

This section describes how to deploy the CyPerf controller and agent on KVM using the QCOW2 images available for download from the [Keysight CyPerf Support](#) page. You can use either *virt-install* or *virt-manager* for the deployment. Use *virt-install* for Ubuntu Server or Ubuntu Desktop, and *virt-manager* for Ubuntu Desktop.

Required Settings on Ubuntu Server

Before running *virt-install*, make sure the KVM hypervisor is properly installed and configured in either Ubuntu 22.04 or Ubuntu 24.04. The following settings on Ubuntu are required:

- Start from Ubuntu. User in our scenario is `kvmuser`.
- Update repositories by running `sudo apt update`.
- Install required KVM packages:
`sudo apt -y install qemu-kvm libvirt-daemon-system libvirt-clients
 bridge-utils qemu-guest-agent virt-manager vim cifs-utils`
- Authorize users to run VMs by adding `kvmuser` to `sudoers` `kvmuser ALL=(ALL) NOPASSWD:ALL`.
- Setup bridge networking with KVM by configuring `br0` in `netplan` with the code below. Then, execute `sudo netplan apply`.

```
# Let NetworkManager manage all devices on this
system
network:
  version: 2
  renderer: NetworkManager
  ethernet:
    ens160:
      dhcp4: yes
  bridge:
    br0:
      dhcp4: yes
  interface:
    - ens160
```

KVM resource requirements

The following hardware resource allocation is recommended for the CyPerf controller and agent:

CyPerf controller	CyPerf agent
8 CPU cores	4 CPU cores
16 GB of RAM	8 GB of RAM
At least 100 GB of SSD storage with thin (i.e. dynamic) provisioning, and 250 GB of SSD storage otherwise	At least 100 GB of SSD storage

NOTE: The CyPerf agent supports **Ubuntu 22.04** and **Ubuntu 24.04** host.

Deploying the CyPerf Controller and Agent on KVM using *virt-install*

You can deploy a CyPerf VM in a KVM environment using the hypervisor's command line interface by running *virt-install*:

1. Download and save the CyPerf controller and agent QCOW2 images to your deployed KVM hypervisor. The CyPerf QCOW2 files are available for download from the [Keysight CyPerf Support](#) page.
2. Then, run the following commands in the hypervisor's terminal:
 - For the **CyPerf controller**:


```
sudo virt-install --connect qemu:///system --virt-type kvm --name mdw --ram 16384 --vcpus=8 --osinfo detect=on,require=off --disk path=/var/lib/libvirt/cyperf-mdw.qcow2,format=qcow2 --import --network bridge=br0,model=virtio --noautoconsole --console pty,target_type=serial
```
 - For the **CyPerf agent**:


```
sudo virt-install --connect qemu:///system --virt-type kvm --name agent1 --ram 8192 --vcpus=4 --osinfo detect=on,require=off --disk path=/var/lib/libvirt/cyperf-agent1.qcow2,format=qcow2 --import --network bridge=br0 --network network=default --noautoconsole --console pty,target_type=serial
```

Where:

- `--name mdw` or `--name agent1` specifies the name of the new virtual machine (VM).
- `--ram 16384` or `--ram 8192` allocates 16 GB RAM or 8 GB RAM respectively to the VM.
- `--vcpus=8` or `--vcpus=4` assigns the required number of CPU cores to the VM.
- `--osinfo detect=on,require=off` allows deployment of the VM without explicitly specifying the OS. KVM will detect it automatically.

- `--disk path=/var/lib/libvirt/cyperf-mdw.qcow2,format=qcow2` or `--disk path=/var/lib/libvirt/cyperf-agent1.qcow2,format=qcow2` specifies the path and format of the QCOW2 file that will be used to create the VM.
- `--import` skips the OS installation, and builds the VM around an existing disk image specified under `--disk path=<QCOW2 image path and name>`.
- `--network bridge=br0,model=virtio` connects to a bridge device in the VM named `br0`, and it specifies the model as `virtio`.
- `--network bridge=br0` connects to a bridge device in the VM named `br0`.
- `--network network=default` connects to a virtual network in the VM named `default`.
- `--noautoconsole` disables the automatic connection to the VM console.

**NOTE:**

- The CyPerf agent can be deployed via a QCOW2 image on hypervisors running either *ARM* or *x86_64 architectures*.
 - **Image format for ARM architecture:** `cyperf-image-1-0-<build>-master-agent-<version>-aarch64.qcow2`
 - **Image format for x86_64 architecture:** `cyperf-image-1-0-<build>-master-agent-<version>-x86-64.qcow2`
- To deploy the CyPerf agent in environments where SSSE3 is not supported on the host hardware, you first need to download the `cyperf_agent_x86_64_old_<stack>_<release/debug>_<x.x.x.x>.deb` from the [Keysight CyPerf Support](#) page, and install it on the agent virtual machine. Use the Debian installation steps described in [Installation Steps](#) under [Commercial Off-the-Shelf \(COTS\) Agent Installation](#).

Deploying CyPerf Controller and Agent on KVM using *virt-manager* GUI

1. Download and save the CyPerf controller and agent QCOW2 images to your deployed KVM hypervisor. The CyPerf QCOW2 files are available for download from the [Keysight CyPerf Support](#) page.
2. Launch Virtual Machine Manager on your KVM environment by selecting **Applications > System Tools > Virtual Machine Manager**, or from the hypervisor's command line by running `sudo virt-manager`. The **Virtual Machine Manager** window opens listing the available VMs already running on the system (if any).
3. Select the **Create a new virtual machine** () button. The **New VM** window opens.
4. Select the **Import existing disk image** option, and then click **Forward**.
5. Click **Browse** to select the CyPerf QCOW2 image to be used for the new VM, and click **Open**. If you saved the CyPerf QCOW2 files to `/var/lib/libvirt/images`, the files will be visible on the right side of the window. If you saved the files in a different location on your server, select **Browse Local**, locate the QCOW2 file to use, and then click **Choose Volume**.

6. Select the **OS type** and **Version** you are running, i.e. **Linux Debian 12**.
7. Enter the memory and number of CPUs to allocate to the VM, as per recommended hardware resource allocation. Then, click **Forward**.
8. Enter a name for the new CyPerf VM, and select the management interface to be used by the VM. Then, click **Finish**.



NOTE:

If CyPerf agents deployed on KVM do not display the login console, send a **Ctrl+Alt+F4** command to bring up the console.

CHAPTER 4

Cloud, Kubernetes, and Container Deployment

This chapter outlines key considerations for deploying each CyPerf component in the cloud, on Kubernetes, and in container environments.

Deploying the CyPerf Controller and Agent in the Cloud, on Kubernetes, and in Containers

Before deploying the templates for your chosen environment, it is recommended to review the architecture, security, and other considerations outlined in this guide. For more information, see the following CyPerf external links:

- [AWS automated deployment](#)
- [Azure automated deployment](#)
- [GCP automated deployment](#)
- [K8s deployment](#)
- [Docker container deployment](#)

Deploying CyPerf in AWS Using the Controller Proxy and Agent Pair

This solution uses a CloudFormation template to deploy the CyPerf controller proxy and two CyPerf agents in an Amazon Virtual Private Cloud solution. Choose this option if you want to deploy the CyPerf controller in a different region or cloud. The CyPerf controller connects to the public IP of the CyPerf controller proxy on port 443 and accesses the CyPerf agents through it. This ensures that all communication between the CyPerf controller and CyPerf agents is encrypted.

You can use the templates in two ways:

- Use a *new* VPC template: This means that all the required network resources are created from scratch (for example, VPC, subnets, route table, Internet gateway, NAT gateway).
- Use an *existing* VPC template: This means that all the required network resources already exist (for example, VPC, subnets, route table, IGW, NAT gateway, security groups). You can select an existing VPC, subnet, and security group during the deployment process.

Each agent has two interfaces:

- The *Management* interface: The agents communicate with the controller proxy through this interface.
- The *Test* interface: Traffic flows through this interface.

During the first deployment, a default agent is set as the Test interface, and the Management interface is set as the second interface. This means that the deployment test traffic flows only between the first interface of both agents.

Troubleshooting AWS Deployment Issues

Q: I encountered a `CREATE_FAILED` error when I launched the Quick Start.

A: If AWS CloudFormation fails to create the stack, it is recommended that you relaunch the template with **Rollback on failure** set to **No**. You can find this setting under **Advanced**, on the **Options** page, in the AWS CloudFormation console. When this setting is set to **No**, the stack's state will be retained, and the instance will be left running, so you can troubleshoot the issue. For details on the error, look for the log files in **%ProgramFiles%\Amazon\EC2ConfigService** and **C:\cfn\log**.



IMPORTANT: When you set **Rollback on failure** to **No**, you will continue to incur AWS charges for this stack. Make sure to delete the stack when you finish troubleshooting.

For additional information, see [Troubleshooting AWS CloudFormation](#) on the AWS website.

Q: I encountered a size limitation error when I deployed the AWS CloudFormation templates.

A: It is recommend that you launch the Quick Start templates from the location that Keysight provided or from another S3 bucket. If you deploy the templates from a local copy on your computer or from a non-S3 location, you might encounter template size limitations when you create the stack.

For more information about AWS CloudFormation limits, see the [AWS documentation](#).

CHAPTER 5

Commercial Off-the-Shelf (COTS) Agent Installation

This chapter briefly describes the Keysight CyPerf agent installer and lists the procedures for how to install, uninstall, update, and switch between different installed versions at run-time, as well as various install and uninstall scenarios and expected behavior.

Compatibility

The Debian installer is currently tested for Debian 12 Server. Installing it on any other flavors of Linux might not work as expected.

Where to Find the Installer

A Debian installer package is provided to users.

Installer Content

The installer is located under the **/opt/keysight/tiger/<build version>/<build configuration>/** directory. Sub-directories are created as follows:

- **bin**: Binaries, including CyPerf agent libraries, AppsecAgent, portmanager, CliCommunicatorUtil
- **cert_key**: Certificates and keys
- **data**: Test input data
- **source**: Common code (data-structures) shared with other components (Middleware)
- **test_config**: CyPerf agent test configuration XML files

Installation Steps

1. Execute the command `sudo apt install cyperf_agent_<architecture>_<stack>_release_<version>.deb` to install the Debian installer.

If any existing *portmanager* configurations are present in the system (such as controller URL, controller proxy URL, Management Interface, or Test Interfaces), they will be reused. If no pre-existing configurations are found, then a new configuration will be used with default configuration values.

2. Use the CLI *cyperfagent* to configure the required configuration parameters. The following commands are available in the CLI:

- `cyperfagent controller show`: Shows the currently configured controller.
- `cyperfagent controller set <Controller-IPv4|Controller-Hostname> <[--interactive]>|<[--username=<USERNAME> --password=<PASSWORD>] [--fingerprint=<FINGERPRINT>]| [--skip-identity-verification]> | [flags]`: Sets the controller.

For more details, see [Connecting CyPerf Agents to a Controller](#).

- `cyperfagent interface management show`: Shows the currently configured Management interface.
- `cyperfagent interface management set ens160`: Sets the Management interface to `ens160`.
- `cyperfagent interface management set auto`: Sets the Management interface to `auto` for dynamically detecting the interface at runtime.
- `cyperfagent interface test set ens160`: Sets the Test interface to `ens160`.
- `cyperfagent interface test set auto`: Sets the Test interface to `auto` for dynamically detecting the interface at runtime.



NOTE:

- The CyPerf Agent can natively run on ARM-based Linux systems using the *aarch64* Debian package. Use the following package format for ARM architecture:
`cyperf_agent_aarch64_ixstack-raw_release_<version>.deb` (for example, `cyperf_agent_aarch64_ixstack-raw_release_26.0.3.834.deb`)
- You cannot install the same version of the application twice on the same system.

Installing an Existing Version

You cannot install the same version twice on the system.

Installing the *ixstack* installer when the `PORT_MANAGER_TEST_INTERFACE` is defined

If the Test interface network driver does not support DPDK, the installation will fail.

How to Uninstall

ⓘ **IMPORTANT:** The uninstall path is not completely implemented yet. To properly complete the uninstall procedure, some manual steps are required. See the [Known Limitations](#) section for details.

To uninstall CyPerf:

1. Use the `apt list --installed | grep cyperf-agent` command to list all the CyPerf agent packages installed on the system.
2. Use `sudo apt remove <cyperf-agent package name>` to uninstall the CyPerf agent version entirely from the system.

If uninstall fails, refer to item 3 of the [Known Limitations](#) section.

Known Limitations

1. The portmanager service is **not** stopped during the installation, but the portmanager which was installed by the current version of the installer is removed.
2. The portmanager configuration file `/etc/portmanager/portmanager-config.json` is not removed during the uninstall process.
3. If the current version is active (i.e., `/opt/keysight/tiger/active` points to it), uninstall will fail. Uninstall is possible after manually removing the active link.

CHAPTER 6

Connecting CyPerf Agents to a Controller

You can connect agents to a controller or controller proxy using either the controller's IPv4 address, IPv6 address, or hostname from its management interface. After the connection succeeds, agents will appear on both the **Agent Management** page and the **Assign Traffic Agents to IP Network** dialog in the CyPerf controller web UI.

① **NOTE:** When an agent has both IPv4 and IPv6 addresses on its management interface, only the IPv4 address will be shown on the **Agent Management** page and the **Assign Traffic Agents to IP Network** dialog.

To connect agents to a controller or controller proxy, use the CLI `cyperfagent set` command:

```
cyperfagent controller set <Controller-IP|Controller-Hostname> <[--
interactive]>|<[--username=<USERNAME> --password=<PASSWORD>] [--
fingerprint=<FINGERPRINT>]|[--skip-identity-verification]> | [flags]
```

Examples:

```
set 192.168.3.4 <params>
set 2001:0db8:85a3:0000:0000:8a2e:0370:7334 <params>
set 192.168.3.4:30422 <params>
set 2001:0db8:85a3:0000:0000:8a2e:0370:7334:30422 <params>
set my-controller.example.com <params>
```

Flags:

<code>--fingerprint string</code>	Sets the controller's key fingerprint parameter.
<code>-h, --help</code>	Displays help for <code>set</code> command.
<code>--interactive</code>	Connects to the controller interactively.
<code>--password string</code>	Sets the controller's authentication password.
<code>--skip-identity-verification</code>	Skips the controller's identity verification.
<code>-s, --skip-restart</code>	Skips <i>portmanager</i> restart.
<code>--username string</code>	Sets the controller's authentication username.

Connect Agents to a Controller with Default Credentials

Without setting the key fingerprint, username, and password parameters, *cyperfagent* will trust the controller fingerprint, and connect using the default controller credentials (i.e., username: `admin`, and password: `CyPerf&Keysight#1`).

For example:

```
cyperfagent controller set 10.38.166.147
```

The terminal displays a message informing you that the default credentials were used to set the controller successfully, and shows that the connection was established.

Connect Agents to a Controller via Command Line Parameters

The controller fingerprint and credentials can be specified via command line parameters.

For example:

```
cyperfagent controller set 10.38.166.147 --username "admin" --password  
"CyPerf&Keysight#1" --  
fingerprint "SHA256:dgBd+IKW5GsN8eXec3f/Mm1XRKQmHvfM73gdcZQDdlU"
```

or

```
cyperfagent controller set 10.38.166.147 --username "admin" --password  
"CyPerf&Keysight#1" --skip-identity-verification
```

The terminal displays a message informing you that the controller is set successfully, and shows that the connection was established.

See also [Manage Server Fingerprints](#).

Connect Agents to a Controller Interactively

You can also use the CLI to connect agents to a controller in an interactive way, by passing the controller fingerprint and credentials when prompted by the terminal. To use the interactive mode, run `cyperfagent controller set` with the `--interactive` parameter. For example:

1. Run `cyperfagent controller set 10.38.166.147 --interactive`.
The terminal prompts you to confirm whether you want to continue.
2. Enter `yes`.
The terminal prompts you to enter the controller's credentials.
3. Enter the default credentials (username: `admin`, password: `CyPerf&Keysight#1`) or the credentials you use to log in the CyPerf controller.
A message is displayed informing you that the controller is set successfully, and shows that the connection was established.

Manage Server Fingerprints

You can copy or revoke a server fingerprint from the **Key Management** dialog in the CyPerf web UI.

To copy a server fingerprint:

1. Log in the CyPerf web UI.
2. Under the **Settings** menu, select **Controller Security > Key Management**. The **Key Management** dialog opens.
3. Copy the key by clicking the **Copy** button () and use it for authentication when connecting the agent to the controller or controller proxy.

To revoke a server fingerprint, click the **Revoke server fingerprint** button, which generates a new key pair.



IMPORTANT: All active connections are terminated when activating the **Revoke server fingerprint** button.

Reset an Agent Configuration

To reset a CyPerf agent's configuration to the factory default settings:

1. Run the following command: `sudo cyperfagent configuration reset hard`
The terminal displays a warning message that all existing agent configurations will be lost, and it prompts you to confirm your action.
2. Enter `yes`.
Progress and status information is displayed for each component, and a success message informs you that the agent was reset successfully.

Troubleshooting the CyPerf Agent Connection

When you try to connect agents to a controller using incorrect credentials or an incorrect fingerprint with the `cyperfagent` command, the connection will not get established, and it will eventually time out with `Connection status: Unauthorized` message. The same message is returned when running the `cyperfagent controller show` command.

The CyPerf controller web UI will also display a banner with a message of the type: `Received authorization requests with incorrect credentials from the following IPs: <agent IP address>`

The banner is persistent and is displayed for all the users of the controller. To resolve these types of errors follow the instructions below.

Incorrect Credentials Errors

- If the incorrect credentials banner is displayed in the controller UI, retry the agent connections using the correct credentials. Use the CLI `cyperfagent set` command, and pass the correct credentials either via the `--fingerprint`, `--username`, and `--password` parameters (see [Connect Agents to a Controller via Command Line Parameters](#)), or via the `--interactive` parameter (see [Connect Agents to a Controller Interactively](#)).
- If running `cyperfagent controller show`, and the connection status message is `Unauthorized`, retry the agent connections using the correct credentials. Use the CLI `cyperfagent set` command, and pass the correct credentials either via the `--fingerprint`, `--username`, and `--password` parameters (see [Connect Agents to a Controller via Command Line Parameters](#)), or via the `--interactive` parameter (see [Connect Agents to a Controller Interactively](#)).

Fingerprint Errors

If running `cyperfagent controller show`, and the connection status message displays a fingerprint error, retry the connection using the correct fingerprint. Use the CLI `cyperfagent set` command, and pass the correct fingerprint either via the `--fingerprint`, `--username`, and `--password` parameters (see [Connect Agents to a Controller via Command Line Parameters](#)), or via the `--interactive` parameter (see [Connect Agents to a Controller Interactively](#)).

Connectivity Errors

If running `cyperfagent controller show`, and the connection status message displays a connectivity error, follow the instructions in [Connecting CyPerf Agents to a Controller](#).

Resetting an Agent Configuration

If connection errors still persist, consider resetting the agent's configuration to the factory default settings. See [Reset an Agent Configuration](#).

CHAPTER 7

Update Software

You can update all the required CyPerf and ATI software components, including the CyPerf agents connected to the current controller, from the **Software Updates** page that is accessible from the **Settings** menu of the CyPerf web UI. You can also update CyPerf agents from the **Agent Management** page as described in *Update Agents from the Controller UI* in the *CyPerfUser Guide*.

- ① **NOTE:**
- An automatic agent update during the CyPerf controller update is available only for **4.0** and newer CyPerf versions, up to CyPerf **6.0**.
 - Upgrade from CyPerf **6.0** to CyPerf **7.0** is *not supported* because of the migration to Debian base images.

To update the required components to a newer version:

1. From the [Keysight CyPerf Support](#) page, download the CyPerf controller upgrade package or the ATI upgrade package, and save it locally.
User registration is required to access the [Keysight CyPerf Support](#) page.
2. Access the CyPerf web UI and log in.
3. Under the **Settings** menu, select **Software Updates**. The **Software Updates** page opens.
4. Click **Select packages for upload**, and choose the upgrade package you downloaded in step 1.
5. Then, click **Open**. The **Upload file** dialog opens informing you of the estimated time it takes to upload the selected package, and warning you against refreshing the page during the upload.
6. Click **Upload** to confirm the action.

- ① **IMPORTANT:** Upload progress will be lost upon page refresh.

The upgrade package starts uploading and progress is shown on the screen. When the upload is complete, a notification is displayed under **Events** informing you that the upload finished successfully. The changes that will be applied are highlighted for each corresponding component.

If an error occurs during the upload process, an **Upload report** will be displayed on the **Software Updates** page.

7. Click **Start update** to initiate the software update process.
Starting from release 4.0, CyPerf agents that are identified not to be running the recommended version will be automatically updated during the CyPerf controller update. When

this happens, a notification appears in the lower right corner of the screen, informing you of the number of agents that require an update and the version to which they will be updated.

For agents that don't support the update feature from the controller UI (specifically, those running on versions older than 3.0), a notification prompts for a manual update. However, the CyPerf controller update continues even if an agent update fails for any reason.



NOTE: These notifications will not appear immediately after initiating the controller update. Only when the deployment service installs the package that checks which agents support the update operation, will these notifications be displayed. Even if you miss these notifications during the update process, you can access them later from the **Events** page.

When the update process successfully completes, the newer software version for each upgraded package will be listed on the **Software Updates** page.



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